

POINTS TO REMEMBER



Remember that data modeling is an iterative process, and you may need to make adjustments based on feedback and changing business requirements. Power BI provides a powerful environment to create robust data models and derive valuable insights from your data.

DATA TUTORIALS
DATA IS THE NEW OIL

DATA TUTORIALS



POINTS TO REMEMBER



6. Data Model Simplification: Keep your data model as simple as possible. Avoid redundant or unnecessary tables and relationships, which can lead to confusion and increased maintenance overhead.

7. Row-level Security (RLS): Implement row-level security to restrict data access based on user roles and permissions. Define appropriate roles and filters to ensure data privacy.

8. Optimize for Performance: Optimize your data model for better performance. Use incremental refresh if dealing with large datasets, and be mindful of the number of visuals and data points in the report.

9. Test Thoroughly: Validate your data model thoroughly to ensure that the relationships, calculations, and visuals are working correctly. Check for data discrepancies and verify the accuracy of key metrics.

10. Documentation: Document your data model, including relationships, calculations, and data sources. This documentation will be valuable for you and other team members who work on the report in the future.

Data Refresh: Set up scheduled data refresh to ensure that your reports always display the most up-to-date information.

11. Version Control: If you work in a team, use version control for your Power BI files to track changes and avoid accidental data model modifications.



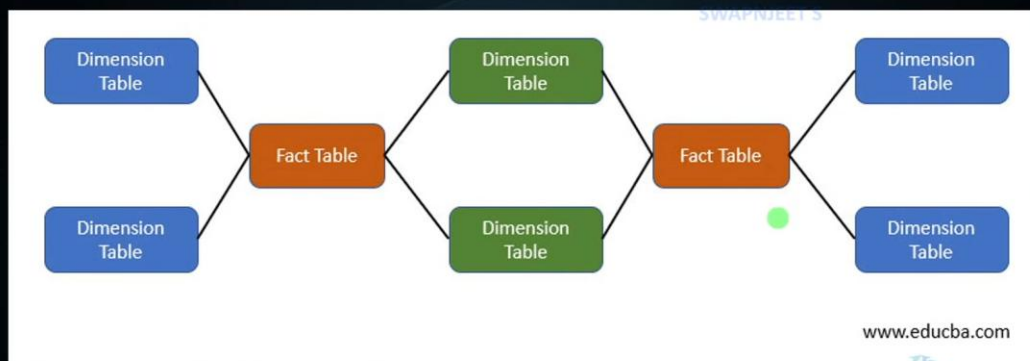
POINTS TO REMEMBER



Creating data modeling in Power BI requires careful consideration and attention to detail to ensure the accuracy, efficiency, and usability of your reports and dashboards

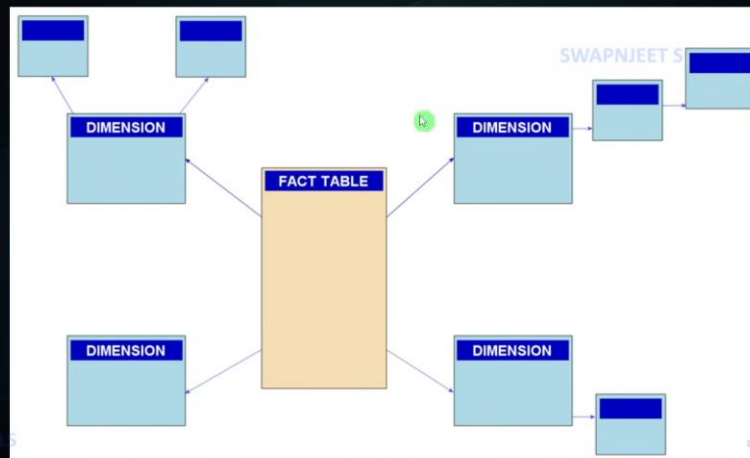
- 1. Data Source Quality:** Ensure that your data source is reliable, clean, and consistent. Check for data quality issues such as missing values, duplicates, and incorrect data types before importing the data into Power BI.
- 2. Data Transformation and Cleanup:** Use Power Query Editor to clean and transform your data. Remove unnecessary columns, handle missing data, and apply data type conversions appropriately. This step is crucial for preparing the data for analysis.
- 3. Relationships:** Establish accurate and meaningful relationships between tables. Check the cardinality between tables to avoid incorrect aggregations and results. Make sure the relationships reflect the actual business logic and join keys.
- 4. Avoid Many-to-Many Relationships:** Try to avoid creating many-to-many relationships as they can lead to data ambiguity and incorrect calculations. Use bridge tables or resolve many-to-many relationships using DAX measures if necessary.
- 5. Use Hierarchies Wisely:** Create hierarchies to enable users to navigate easily through data, but avoid creating too many levels in a hierarchy, as it may overwhelm users and make the report cluttered.

GALAXY SCHEMA



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1.Active Relationship:

1. An active relationship is the default relationship that Power BI uses when querying the data. It is the primary relationship that determines how tables are related and how data is aggregated in visuals and calculations.
2. When you create multiple relationships between two tables, Power BI automatically identifies one of them as the active relationship based on the data model's cardinality and the direction of the relationships.
3. The active relationship is used for standard calculations, such as summing values, filtering data, and performing aggregations in visuals and measures.

2.Inactive Relationship:

1. An inactive relationship is an additional relationship that is not used by default when querying the data. It is intended to be used selectively, depending on the analysis requirements or user interactions.
2. The inactive relationship does not affect standard calculations or visuals directly. Instead, it allows you to create custom calculations using the `USERELATIONSHIP` function in DAX.
3. The `USERELATIONSHIP` function enables you to override the active relationship and specify which relationship to use for a specific calculation or visualization. This gives you more control over how data is related and displayed in

CARDINALITIES IN POWER BI



1. One-to-One (1:1): In a one-to-one relationship, each record in the first table is related to only one record in the second table, and vice versa. This type of relationship is relatively rare in data modeling but can be useful in certain situations where you need to split large tables for performance reasons or when you have optional attributes that can be stored in a separate table.

2. One-to-Many (1:N): The one-to-many relationship is the most common type of relationship in data modeling. In this relationship, each record in the first table can be related to multiple records in the second table, but each record in the second table is related to only one record in the first table. This is the standard relationship type used to connect dimension tables to fact tables in a star schema.

3. Many-to-Many (N:N): Many-to-many relationships are less common and are usually resolved using bridge tables. In a many-to-many relationship, each record in the first table can be related to multiple records in the second table, and vice versa. This type of relationship can create data ambiguity, so it is essential to use bridge tables or DAX measures to resolve the relationship and perform accurate calculations.



FACT & DIMENSION TABLE



Dimension Table:

- Also called as **Lookup Table** or **Mapping Table**
- Dimension tables contain descriptive attributes that provide context to the measures in the fact table. They provide a way to categorize, filter, and group the data in the fact table.
- Dimension tables typically have fewer rows compared to fact tables and store attributes like customer names, product categories, geographic locations, time periods, and other qualitative information.
- Dimension tables are used for slicing and dicing the data, providing meaningful labels to the data points in the fact table, and offering the ability to drill down into more detailed information.
- A Dimension Table contains Primary key which creates a relationship with Fact table Foreign key

Example: In the e-commerce scenario, dimension tables could include attributes like CustomerID, CustomerName, ProductID, ProductName, Category, Subcategory, LocationID, LocationName, Date, and so on.



FACT & DIMENSION TABLE



Fact Table:

- Also Called as **Data or Master Table**
- A fact table contains the quantitative data that represents business transactions or events. It typically holds the numeric measures or metrics that you want to analyze, such as sales revenue, quantity sold, profit, or expenses.
- Each row in the fact table corresponds to a specific event or transaction and includes foreign keys that link to the related dimension tables. These foreign keys represent the context of the event and allow you to slice and dice the data along different dimensions.
- Fact tables are often large in size and can grow rapidly over time, as they store detailed data at the most granular level.

Example: Consider an e-commerce business. The fact table could contain information about each individual purchase, such as the product sold, the quantity purchased, the sales amount, the date of purchase, and foreign keys pointing to the related dimension tables like Customer, Product, and Time.

WAYS TO CREATE RELATIONSHIP



1. Autodetect New Relationship after Data is loaded : The relationship is auto detected by Power BI based on same column names and data type in two tables
2. Drag and Drop Columns from two tables you want to have relationship between.
3. Edit Relationship Tab Option – Select two tables from dropdown list and then select the columns you want to have relations between

RAW DATA

Table Name	Table Type
orders_dataset	Fact Table
customers_dataset	Dimension Table
delivery_status_dataset	Dimension Table
payment_mode_dataset	Dimension Table
products_dataset	Dimension Table
sellers_dataset	Dimension Table
reviews_dataset	Dimension Table



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DATA MODELING

Data modeling in Power BI is a crucial step in building effective and insightful reports and dashboards. It involves shaping and transforming your data to create a structured and optimized data model that supports the visualization and analysis requirements of your business

The purpose of data modeling in Power BI is to create a structured, optimized, and meaningful representation of your data to support effective analysis, reporting, and visualization. Data modeling is a critical step in the Power BI workflow that transforms raw data into actionable insights for decision-making and business intelligence

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WHAT WILL YOU LEARN



- How to Create Data Model for Real time Data Tables
- Fact and Dimension Table
- Ways to Create Relationship in Power BI
- How to identify Tables with Foreign and Primary Keys
- Cardinalities in Power BI
- Active and Inactive Relationships with Example
- Star, Snowflake and Galaxy Schemas
- Important Points to Remember while creating Data Model

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STEPS TO REMEMBER WHILE CLEANING

Cleaning data effectively requires attention to detail and a systematic approach. Here are some important points to remember while cleaning data:



- | | |
|------------------------------------|-----------------------------|
| ✓ Understand Your Data | ✓ Test Data Transformation |
| ✓ Focus on Domain Knowledge | ✓ Iterative Process |
| ✓ Create a Backup | ✓ Address Data Entry Errors |
| ✓ Document Your Process | ✓ Test Rigorously |
| ✓ Prioritize Issues | ✓ Consider Data Privacy |
| ✓ Standardize Formats (Data Types) | ✓ Automate When Possible |
| ✓ Verify Data Integrity | ✓ Communicate Findings |
| ✓ Use Descriptive Statistics | ✓ Validate After Cleaning |

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IF DATA USED IN NOT CLEANED

Failing to clean your data before analysis can have several negative effects that can undermine the accuracy, reliability, and meaningfulness of your results. Here are some consequences of not cleaning your data

- Inaccurate Insights
- Failed Data Models
- Misleading Visualizations
- False Conclusions
- Loss of Credibility
- Wasted Resources
- Legal and Compliance Issues
- Reputational Damage
- Difficulty in Integration
- Increased Risk



WHY TO CLEAN DATA ?

Data collected from various sources can be messy, containing errors due to human mistakes, technical issues, or other factors. Data cleaning involves various methods and techniques to transform the data into a more accurate, consistent, and usable format

- ✓ Ensuring Data Accuracy
- ✓ Enhancing Data Quality
- ✓ Preventing Inaccurate Insights
- ✓ Supporting Reliable Analysis
- ✓ Maintaining Data Consistency
- ✓ Reducing Data Processing Time
- ✓ Minimizing Errors in Models and Algorithms
- ✓ Optimizing Resources
- ✓ Improving Decision-Making
- ✓ Enhancing Data Collaboration

